Umberto Tarantino

76 rue du Cardinal Lemoine
75005 Paris, France
+39 345 928 19 49
umbertotarantino17@gmail.com
he/they

Research interests

Main — Categorical logic, topos theory, duality theory
Others — Intuitionistic logic, model theory, realizability, categorical semantics

Education

Oct 2024 – present — PhD in Theoretical Computer Science · IRIF, Université Paris Cité
Thesis: Classifying topoi and intuitionistic model theory through ultracategories and polyadic spaces
Advisor: Sam van Gool

Sept 2023 – July 2024 — Erasmus+ · Universiteit van Amsterdam

Sept 2022 – July 2024 — MSc in Mathematics · Università degli Studi di Milano
Thesis: A category of arrow algebras for modified realizability
Advisors: Benno van den Berg and Silvio Ghilardi
110/110 cum laude

Sept 2019 – July 2022 — BSc in Mathematics · Università degli Studi di Milano
110/110 cum laude

Publications

2026 — *Toposes with enough points as categories of étale spaces*
Joint work with Sam van Gool and Jérémie Marquès
Journal of Pure and Applied Algebra, 230 (6): 108269

2025 — *A category of arrow algebras for modified realizability*
Theory and Applications of Categories, 44 (4): 132–180

Preprints

2026 — *Categorified convergence: dropping the 'ultra' from 'virtual ultracategories'*
Joint work with Gabriel Saadia

Logics vs doctrines, a mirror symmetry
Joint work with Ivan Di Liberti and Lingyuan Ye

Conceptual completeness for subgeometric logics
Joint work with Ivan Di Liberti and Lingyuan Ye

Profunctorial algebras
Joint work with Quentin Aristote

2025 — *Ultracategories via Kan extensions of relative monads*
Joint work with Joshua Wrigley

Teaching

Fall 2026 — Logique · L3 Informatique, *Université Paris Cité*
Exercise classes (in French) · Lecturer: Sylvain Schmitz

Spring 2026 — Introduction à la programmation 2 · L1 Informatique, *Université Paris Cité*
Lab classes (in French) · Lecturer: Daniela Petrişan

Fall 2025 Programmation 1 · S5 EIDD, *Université Paris Cité*
Lab classes (in French) · Lecturer: Pierre Charbit

Fall 2024 Programmation 1 · S5 EIDD, *Université Paris Cité*
Lab classes · Lecturer: Pierre Charbit

Spring 2024 Topos theory · MasterMath, *Utrecht University*
Exercise classes · Lecturer: Jaap van Oosten

Community and events

2026 – present Curator of the *ItaCa* website

2025 – present Organizer of the *Realizability reading group* at IRIF

2025 – present Organizer of the *queeRIF* meetings

Dec 2026 Organizer of the *7th ItaCa Workshop*

June 2026 Organizer of the *Ultracategories and Stone Duality Workshop*

2024 – 2025 Organizer of the *Topos theory reading group* at IRIF

Talks

2026 *An arrow-algebraic approach to realizability*
CHoCoLa · *ENS de Lyon* · invited

Conceptual completeness for geometric logic via ultraconvergence spaces
XXIX Incontro di Logica AILA 2026 · *Rome*

Conceptual completeness for geometric logic via ultraconvergence spaces
TACL 2026 · *Krakow*

Conceptual completeness for geometric logic via ultraconvergence spaces
Making Categorical Model Theory Accessible · *Masarykova Univerzita*

Toposes with enough points as categories of étale spaces
Gothenburg Logic Seminar · *Göteborgs Universitet* · invited

2-dimensional topologies via profunctorial algebras
Stockholm Logic Seminar · *Stockholms Universitet* · invited

Toposes with enough points as categories of étale spaces
7th Workshop on Formal Topology · *Venice*

Toposes with enough points as categories of étale spaces
Séminaires LoCal · *LIPN* · invited

2025 *Toposes with enough points as categories of étale spaces*
ItaCa Workshop 2025 · *Università degli Studi di Milano*

Ultracategories via Kan extensions of relative monads
ItaCa Fest 2025 · *online* · invited

Where do ultracategories come from?
CT2025 · *Masarykova Univerzita*

Arrow algebras, algebraic structures for modified realizability
Realizability Workshop · *CIRM, Marseille*

Arrow algebras, algebraic structures for modified realizability
GT Sémantique · *IRIF*

A gentle introduction to categorical realizability
Séminaire des membres non-permanents · *IRIF*

2024 *Triposes and toposes via arrow algebras*

Categorical Logic and Higher Categories · *University of Manchester*

Triposes and toposes through arrow algebras
Dutch Categories And Types Seminar · *Universiteit Leiden*

Awards and scholarships

June 2025 — AILA 3+2 Award
Best Master's Thesis Award of the Associazione Italiana di Logica e sue Applicazioni

Oct 2024 — Contrat International IDEX
Excellence scholarship for international PhD students at Université Paris Cité

Languages

Italian	native
English	fluent, CEFR C1
French	intermediate

Technical skills

Programming	C, Java, F#
Other	LaTeX, Z3, SPASS